

NEUROSCI 366

Lecture 26

Hopfield 联想记忆模型 / Hopfield Associative Memory

Source-aligned · Static · Searchable · Printable

Source files

- Lecture26-Memory-HopfieldModel.pdf

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：Hebbian assembly。
- 从第一原则推导：binary asynchronous dynamics。
- 解释几何/计算直觉：energy。
- 连接生物学解释：stored attractors/basins。
- 诊断 assumptions 与 failure modes：spurious mixtures。
- 把概念迁移到新神经科学问题：signal+crosstalk。

Dependency map

Hebbian assembly → binary asynchronous dynamics → energy → stored attractors/basins → spurious mixtures → signal+crosstalk

Core question

symmetric Hebbian recurrent weights 如何把 random binary patterns 变成 attractors, energy descent、crosstalk noise 与 loading α 又如何决定 pattern completion 和 capacity?

3 Five-minute prerequisite diagnostic

1. asynchronous update 是什么？（卡住时先看 Source-page unit 1，再看补全推导。）
2. energy descent 需要哪些 weight 条件？（卡住时先看 Source-page unit 2，再看补全推导。）
3. 为什么 $\Delta E \leq 0$ ？（卡住时先看 Source-page unit 3，再看补全推导。）
4. Hebbian W_{ij} 存了什么？（卡住时先看 Source-page unit 4，再看补全推导。）
5. $P=1$ 时 ξ 为什么 fixed point？（卡住时先看 Source-page unit 5，再看补全推导。）

Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

4 Source-aligned lesson / 按原笔记页序

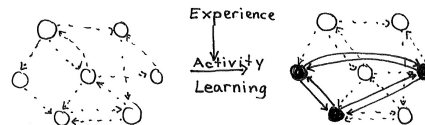
4.1 Lecture26-Memory-HopfieldModel.pdf • p. 1

Explicit memory:

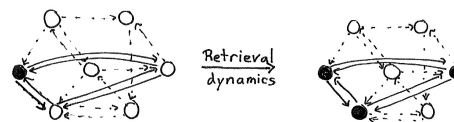
Everyone can think back to recent lectures and recall some things that we covered. How could the brain recall this information?

Hebbian assemblies:

During memory encoding, connections are formed between active neurons.



Recall could subsequently occur by activating a subset of these previously active neurons.



Since retrieval dynamics recreate the same pattern of neural activity present during encoding, you can relive the experience, extract content from it, and so on. In rodents, reactivating hippocampal neurons active during fear conditioning leads to freezing behavior (memory engram).

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 1]

Clean reconstruction. Page 1: explicit memory, Hebbian assemblies, encoding/retrieval, pattern completion, engram.

What the note is saying. 本页的中心对象是 Hebbian assembly 与 binary asynchronous dynamics。原笔记将它们放进以下链条：explicit memory、Hebbian assemblies、encoding/retrieval、pattern completion、engram。

Why it follows. 从 Hebbian cell assembly 解释 encoding、partial cue、retrieval、pattern completion 和 attractor basin；区分 biological hypothesis 与 abstract binary model。其逻辑后果是：从 Hebbian weight rule 完整推导 $P=1$ 时 stored pattern 为 attractor，并分析 aligned/anti-aligned/spurious states。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

asynchronous update 是什么？

4.2 Lecture26-Memory-HopfieldModel.pdf • p. 2

Hopfield model:

Hopfield built an explicit neural network with these properties. It was based on the physics of magnetic systems, so its formulation is a bit different than what we're used to. Physics Nobel in 2024!

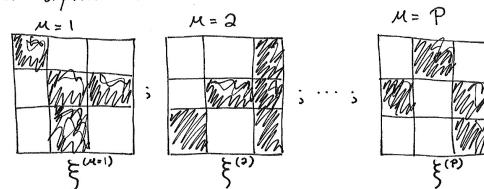
Network dynamics:

- Choose a random neuron, i , for update.
- Update state of neuron by

$$S_i(t+1) = \underbrace{\text{sgn}}_{\pm 1} \left(\sum_{j=1}^N \underbrace{W_{ij}}_{\text{synaptic weights}} S_j(t) \right)$$

Binary state of neuron
- Iterate until convergence.

Goal: Store P patterns as attractors of network dynamics. (NOTE: This depicts 9 neurons in 2D).



$$\mathcal{P}(\xi_i^{(\mu)} = 1) = \mathcal{P}(\xi_i^{(\mu)} = -1) = 1/2 \Rightarrow \langle \xi_i^{(\mu)} \rangle = 0$$

$$\mathcal{P}(\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(P)}) = \prod_{\mu=1}^P \prod_{i=1}^N \mathcal{P}(\xi_i^{(\mu)})$$

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 2]

Clean reconstruction. Page 2: Hopfield binary network, asynchronous update, stored random patterns.

What the note is saying. 本页的中心对象是 energy 与 stored attractors/basins。原笔记将它们放进以下链条：Hopfield binary network、asynchronous update、stored random patterns。

Why it follows. 精确定义 binary states、random patterns、asynchronous single-neuron update 和 convergence procedure。其逻辑后果是：从 Hebbian cell assembly 解释 encoding、partial cue、retrieval、pattern completion 和 attractor basin；区分 biological hypothesis 与 abstract binary model。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 逐条追踪箭头方向、兴奋/抑制符号、状态变量与外部输入；结构图说明允许的依赖关系，不自动证明生物实现。

Stop & Predict

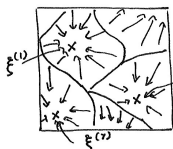
energy descent 需要哪些 weight 条件?

4.3 Lecture26-Memory-HopfieldModel.pdf • p. 3

"Ising Model"

$$E = -\frac{1}{2} \sum_i \sum_{j \neq i} W_{ij} S_i S_j$$

Dynamics minimize energy function



→ : Dynamics from model.
 x : Attractor state, $\xi^{(\mu)}$
 - : Basins of attraction

Strategy: Set synaptic weights symmetrically by Hebbian scheme.

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \underbrace{\xi_i^{(\mu)} \xi_j^{(\mu)}}_{\substack{\text{"1" if same} \\ \text{"-1" if different}}}, i \neq j$$

$$W_{ii} = 0$$

P=1 example:

$$\begin{aligned} S_i(t+1) &= \text{sgn} \left(\sum_{j=1}^N W_{ij} S_j(t) \right) \\ &= \text{sgn} \left(\frac{1}{N} \sum_{j \neq i} \xi_i^{(\mu)} \xi_j^{(\mu)} S_j(t) \right) \quad [u \cdot v = \sum_{j \neq i} u_j v_j] \\ &= \text{sgn} \left(\xi_i^{(\mu)} \left(\xi_j^{(\mu)} : S(t) \right) \right) \\ &= \underbrace{\text{sgn} \left(\xi_i^{(\mu)} \right)}_{\xi_i^{(\mu)}} \underbrace{\text{sgn} \left(\xi_j^{(\mu)} : S(t) \right)}_{\text{If } +1, S_i(t+1) = \text{sgn}(\xi_i^{(\mu)}), \text{ and dot product gets bigger. Memory retrieval!}} \end{aligned}$$

If -1, $S_i(t+1) = -\text{sgn}(\xi_i^{(\mu)}) = -\xi_i^{(\mu)}$, and dot product gets smaller.
 "Spurious attractor" at $-\xi^{(\mu)}$!

Starting in $\pm \xi^{(\mu)}$, the system stays there.

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 3]

Clean reconstruction. Page 3: Ising energy, symmetric Hebbian weights, P=1 retrieval derivation, basins/spurious attractors.

What the note is saying. 本页的中心对象是 spurious mixtures 与 signal+crosstalk。原笔记将它们放进以下链条：Ising energy、symmetric Hebbian weights、P=1 retrieval derivation、basins/spurious attractors。

Why it follows. 解释 Ising energy, 证明 symmetric weights/asynchronous updates 使 energy non-increasing; 说明 synchronous update 的差异。其逻辑后果是：从 Hebbian cell assembly 解释 encoding、partial cue、retrieval、pattern completion 和 attractor basin; 区分 biological hypothesis 与 abstract binary model。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

为什么 $\Delta E \leq 0$?

4.4 Lecture26-Memory-HopfieldModel.pdf • p. 4

In general, the original Hopfield model has spurious attractors at odd mixture states

$$\begin{aligned} 1\text{-mixture: } & \text{sgn}(\pm \xi^{(u)}) \\ 3\text{-mixture: } & \text{sgn}(\pm \xi^{(u_1)} \pm \xi^{(u_2)} \pm \xi^{(u_3)}) \\ & \vdots \end{aligned}$$

Memory capacity: How many random patterns can be stored as attractors? Suppose $s(t) = \xi^{(u)}$. Does it stay there?

$$\begin{aligned} s_i(t+1) &= \text{sgn}\left(\sum_{j=1}^N W_{ij} \xi_j^{(u)}\right) = \text{sgn}\left(\frac{1}{P} \sum_{j \neq i} \sum_{\nu \neq u} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(u)}\right) \\ &= \text{sgn}\left(\underbrace{\frac{N-1}{N} \xi_i^{(u)}}_{\text{Signal from } u} + \underbrace{\frac{1}{N} \sum_{j \neq i} \sum_{\nu \neq u} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(u)}}_{\text{Interference from } \nu \neq u}\right) \\ &= \text{sgn}\left(\frac{N-1}{N} \xi_i^{(u)} + \frac{1}{N} \eta_i^{(u)}\right) \end{aligned}$$

$$\langle \eta_i^{(u)} \rangle = 0$$

$$\begin{aligned} \langle (\eta_i^{(u)})^2 \rangle &= \left\langle \sum_{j \neq i} \sum_{\nu \neq u} \sum_{K \neq i} \sum_{g \neq u} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(u)} \xi_K^{(g)} \xi_i^{(g)} \xi_K^{(u)} \right\rangle \\ &= \sum_{j \neq i, K \neq i, \nu \neq u, g \neq u} \underbrace{\langle \xi_i^{(\nu)} \xi_i^{(g)} \rangle}_{\delta_{\nu g}} \underbrace{\langle \xi_j^{(\nu)} \xi_K^{(u)} \rangle}_{\delta_{jK}} \\ &= \underbrace{\langle \xi_j^{(\nu)} \xi_K^{(u)} \rangle}_{1 \text{ (b.c. of other 6 terms)}} \end{aligned}$$

$$= (N-1)(P-1)$$

Define $h_i^{(u)} = \sum_{j=1}^N W_{ij} \xi_j^{(u)}$. The picture is

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 4]

Clean reconstruction. Page 4: odd-mixture states, memory capacity, signal-plus-interference decomposition.

What the note is saying. 本页的中心对象是 capacity/spin glass 与 sparse alternative。原笔记将它们放进以下链条: odd-mixture states、memory capacity、signal-plus-interference decomposition。

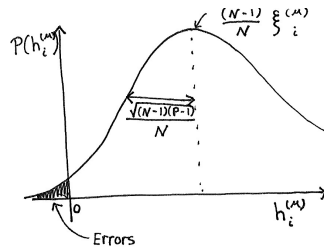
Why it follows. 从 Hebbian weight rule 完整推导 $P=1$ 时 stored pattern 为 attractor, 并分析 aligned/anti-aligned/spurious states。其逻辑后果是: 解释 odd-mixture spurious attractors 及 basin of attraction。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

Hebbian W_{ij} 存了什么?

4.5 Lecture26-Memory-HopfieldModel.pdf • p. 5



As $N \rightarrow \infty$, $\text{mean} \rightarrow \xi_i^{(u)}$, $\sigma \rightarrow \sqrt{\frac{P-1}{N}}$... Substantial errors start to occur when

$$P = \propto N,$$

so the memory capacity is proportional to the number of neurons.

Amit, Gutfreund, and Sompolinsky leveraged their knowledge of disordered magnetic systems to rigorously derive the proportionality constant as

$$\alpha_c = 0.138$$

Below $P_c = \alpha_c N$, there were attractor states differing from the desired patterns by at most 1.5%. Above P_c , the attractor states were uncorrelated to the patterns. This abrupt change is due to a "spin glass" phase transition.

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 5]

Clean reconstruction. Page 5: Gaussian approximation, error probability, $P \propto N$, critical $\alpha \approx 0.138$, spin-glass transition.

What the note is saying. 本页的中心对象是 Hebbian assembly 与 binary asynchronous dynamics。原笔记将它们放进以下链条：Gaussian approximation、error probability、 $P \propto N$ 、critical $\alpha \approx 0.138$ 、spin-glass transition。

Why it follows. 解释 capacity proportional to N 、loading $\alpha = P/N$ 、critical $\alpha_c \approx 0.138$ 和 spin-glass transition；说明该常数依赖经典模型假设。其逻辑后果是：对 $P > 1$ 做 signal-plus-crosstalk decomposition；逐步计算 interference mean/variance，并用 Gaussian approximation 得到 error tendency。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 把图与矩阵 shape 对齐：轴代表变量或 mode，椭圆方向对应 eigenvectors，轴长/曲率对应 eigenvalues；不要把统计轴自动解释为因果通路。

Stop & Predict

$P=1$ 时 ξ 为什么 fixed point?

4.6 Lecture26-Memory-HopfieldModel.pdf • p. 6

Other associative memory models:

Hopfield opened the floodgates, and many physicists started to study neural network models of memory. These models sometimes had better properties than Hopfield's. My favorite is due to Buhmann, Dirko, and Schulten. It is

- $S_i(t) \in \{0, 1\}$, (active vs. inactive)
- $P(\xi_i^{(\mu)} = 1) = \alpha$, (sparse patterns)
- $W_{ij} = \underbrace{\frac{1}{\alpha(1-\alpha)N} \sum_{\mu=1}^P (\xi_i^{(\mu)} - \alpha)(\xi_j^{(\mu)} - \alpha)}_{\text{Hebbian}} - \underbrace{\frac{\gamma}{\alpha N}}_{\text{Global inhibition}}$
- $P(S_i(t+1) = 1) = \frac{1}{1 + e^{-(h_i - u)/T}}$ (stochastic update)

Its nice properties include a parameter regime where

- Attractor at $S(t) = 0$.
- Only other attractors at $\xi^{(\mu)}$.
- Memory capacity and information storage are near optimal for small α .

Original-note anchor. [Source: Lecture26-Memory-HopfieldModel.pdf, p. 6]

Clean reconstruction. Page 6: sparse/stochastic associative-memory alternative, global inhibition, near-optimal capacity.

What the note is saying. 本页的中心对象是 energy 与 stored attractors/basins。原笔记将它们放进以下链条：sparse/stochastic associative-memory alternative、global inhibition、near-optimal capacity。

Why it follows. 比较 sparse $\{0, 1\}$ stochastic model、global inhibition、temperature 与 information capacity。其逻辑后果是：解释 capacity proportional to N 、loading $\alpha = P/N$ 、critical $\alpha_c \approx 0.138$ 和 spin-glass transition；说明该常数依赖经典模型假设。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

- ξ 为什么也 attractor?

5 补全推导与数学支架 / Derivations

5.1 binary asynchronous update 与 energy descent

【必要前置】 为什么 dynamics 收敛而不是任意循环？

$$S_i \leftarrow \text{sgn} \left(\sum_j W_{ij} S_j \right)$$

$$E = -\frac{1}{2} \sum_{ij} W_{ij} S_i S_j$$

$$\Delta E = -(S_i^{\text{new}} - S_i^{\text{old}}) h_i \leq 0$$

1. 每步只随机选择一个 neuron 更新。
2. symmetric $W_{ij}=W_{ji}$ 、 $W_{ii}=0$ 让涉及 i 的 energy change 合并成 $-\Delta S_i h_i$ 。
3. update 选择与 local field h_i 同号的新 state, 因此 product 非负、 $\Delta E \leq 0$ 。
4. 有限 state space 下 eventually 到 fixed point (或平能 plateau)。

Geometric/computational intuition. 每个 bit 局部对齐于 neighbors, 同时整个 network 沿 energy landscape 下坡。

Biological interpretation / failure condition. synchronous update 可出现 2-cycles; asymmetric weights 不保证该 energy。

Micro-check

Hebbian W_{ij} 存了什么？

5.2 Hebbian weights 与 P=1 retrieval

【补全推导】 证明单一 pattern ξ 是 attractor。

$$W_{ij} = \frac{1}{N} \xi_i \xi_j \quad (i \neq j)$$

$$h_i = \sum_{j \neq i} W_{ij} S_j = \frac{\xi_i}{N} \sum_{j \neq i} \xi_j S_j$$

$$S_i^{new} = \xi_i \operatorname{sgn}(\xi \cdot S - \xi_i S_i)$$

1. 若 $S=\xi$, local field $= (N-1)\xi_i/N$, update 保持 ξ_i 。
2. 若 cue 与 ξ overlap 为正且 errors 不多, sum 仍与 ξ_i 同号, errors 被纠正。
3. $-\xi$ 也成为 attractor, 因为 weights 对 global sign invariant。
4. basin size 由 initial overlap 与 finite-size effects 决定。

Geometric/computational intuition. 每条 weight 存储两个 bits 是否同号, local field 投票恢复一致 pattern。

Biological interpretation / failure condition. patterns correlated 或 biased activity 会改变 weights 与 basins。

Micro-check

P=1 时 ξ 为什么 fixed point?

5.3 $P > 1$ 的 signal-crosstalk decomposition

【跨页连接】 在目标 memory μ 上计算 local field。

$$W_{ij} = \frac{1}{N} \sum_{\nu=1}^P \xi_i^\nu \xi_j^\nu$$

$$h_i^\mu = \frac{N-1}{N} \xi_i^\mu + \eta_i^\mu$$

$$E[\eta] = 0, \quad \text{Var}(\eta) \approx \frac{(N-1)(P-1)}{N^2} \approx \alpha$$

1. $v=\mu$ 项给 aligned signal。
2. $v \neq \mu$ memories 的随机 products 相加成 interference。
3. 独立 random ± 1 patterns 下 mean 0, variance 是项数/ N^2 。
4. large N 可用 Gaussian approximation 估 error $P(h_i^\mu < 0)$ 。

Geometric/computational intuition. 每个 memory 都贡献一个 weak vote, 目标 signal $O(1)$, 其他 memories 汇成 noise $O(\sqrt{\alpha})$ 。

Biological interpretation / failure condition. independence/Gaussian approximation 在 correlated/sparse patterns 下改变。

Micro-check

$-\xi$ 为什么也 attractor?

5.4 capacity、spurious mixtures 与 sparse alternative

【模型边界】 为什么 capacity $\propto N$ 且经典常数约 0.138?

$$\alpha = P/N$$

$$\alpha_c \approx 0.138 \quad (\text{classic dense random Hopfield})$$

$$S^{mix} = \text{sgn}(\pm \xi^{\mu_1} \pm \xi^{\mu_2} \pm \xi^{\mu_3})$$

1. 保持 finite error 需 P 与 N 同阶, 因此 capacity linear in N。
2. self-consistent statistical-mechanics analysis 给稳定 retrieval 临界 $\alpha_c \approx 0.138$ 。
3. odd mixtures 可形成 spurious attractors, 不对应任何单一 memory。
4. sparse {0,1} stochastic models 加 global inhibition/temperature, 可提高 information efficiency 并改变 capacity。

Geometric/computational intuition. 更多 memories 让 landscape 出现错误 valleys, 最终 retrieval states 被 spin-glass-like states 取代。

Biological interpretation / failure condition. 0.138N 不是 brain memory capacity, 也不适用于 modern/sparse/structured associative memories。

Micro-check

overlap $m=1, 0, -1$ 分别表示?

6 Cross-page synthesis / 跨页因果与数学链

Hopfield model 把 explicit memory 抽象为 attractor dynamics: partial cue 经 recurrent updates 收敛到 stored pattern。symmetric weights 与 asynchronous single-neuron updates 保证 Ising energy 不增加。P=1 时 Hebbian weights 使目标及其反号成为 fixed points; P>1 时其他 memories 产生 crosstalk noise, loading $\alpha=P/N$ 增大后 retrieval errors 上升, 并在经典 dense random model 的 $\alpha \approx 0.138$ 附近失去稳定 retrieval。

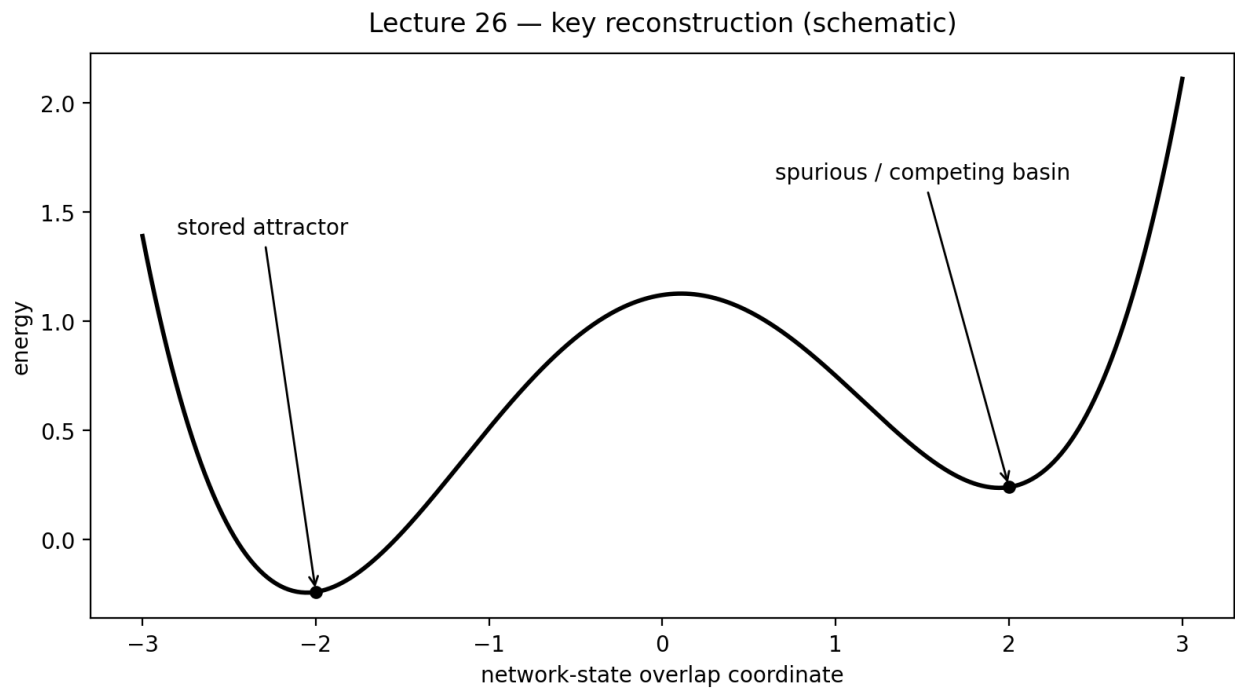


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

7 Worked examples and transfer

7.1 小 N pattern completion

【原课/核心例题】 Prompt. $\xi=(1,1,-1,-1)$, $P=1$; cue $S=(1,-1,-1,-1)$ 。

Worked solution. overlap $\xi \cdot S = 2 > 0$; 对第二 bit, 多数 other aligned products 推 local field 为正, 更新可纠正到 +1, 随后收敛 ξ 。

Check. 需逐 neuron asynchronous 更新; order 可影响 transient, 但 energy 不增。

7.2 loading scale

【跨课连接】 Prompt. $N=1000$, $P=100$ 。

Worked solution. $\alpha=0.1$, 低于经典 0.138 threshold, 理论上 random patterns 可较稳定 retrieval; 这不是保证每个 finite pattern 零错误。

Check. critical value 是 large-N 经典模型结果。

Micro-check

Explain in your own words. 用不超过 120 字解释: 本课从 source page 1 到最后一页, 究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction?

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error? 写出一个具体例子。

8 Formula and notation sheet

1. binary neuron state

$$S_i \in \{-1, +1\}$$

Conditions / conventions: classic dense model

2. asynchronous update

$$S_i \leftarrow \text{sgn}(h_i)$$

Conditions / conventions: one neuron at a time

3. Hopfield energy

$$E = -\frac{1}{2} \sum_{ij} W_{ij} S_i S_j$$

Conditions / conventions: symmetric W

4. Hebbian storage rule

$$W_{ij} = N^{-1} \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}$$

Conditions / conventions: $W_{ii}=0$

5. pattern overlap

$$m^{\mu} = N^{-1} \sum_i \xi_i^{\mu} S_i$$

Conditions / conventions: ± 1 patterns

6. retrieval field decomposition

$$h = \text{signal} + \text{crosstalk}$$

Conditions / conventions: random-pattern analysis

7. loading ratio

$$\alpha = P/N$$

Conditions / conventions: P memories, N neurons

8. classic capacity threshold

$$\alpha_c \approx 0.138$$

Conditions / conventions: dense unbiased random Hopfield

9 Bilingual glossary

中文	English / symbol	Operational meaning
联想记忆 细胞集合	associative memory cell assembly	由 partial/noisy cue 恢复 stored pattern。 共同活动并因 Hebbian connections 形成的 neuron group。
吸引子盆地 模式补全 异步更新 Ising 能量	basin of attraction pattern completion asynchronous update Ising energy	收敛到同一 attractor 的 initial states 集合。 由 incomplete cue 重建完整 memory。 每步只更新一个 unit。 symmetric binary network 的 Lyapunov function。

中文	English / symbol	Operational meaning
重叠	overlap	current state 与 stored pattern 的 normalized dot product.
串扰	crosstalk	其他 stored patterns 对目标 retrieval 的 interference.
伪吸引子	spurious attractor	不对应单一 stored pattern 的 stable state.
自旋玻璃	spin glass	大量无序 metastable states 主导的 regime.
温度	temperature	stochastic update noise 的 scale parameter.

10 Common traps, assumptions, and limitations

1. synchronous 与 asynchronous dynamics 混用 energy proof.
2. 忘记 W 必须 symmetric 且 $W_{ii}=0$.
3. $P=1$ 时只说 ξ attractor, 忽略 $-\xi$ 与 spurious/global symmetry.
4. 把 overlap 与 exact bit accuracy 混为一谈.
5. crosstalk variance 不检查 N 、 P scaling.
6. 把 Gaussian approximation 当精确 finite- N law.
7. 把 $0.138N$ 当作大脑 memory capacity.

Course-specific risk boundary. 明确 “memory capacity $0.138N$ ” 只属于经典 dense random-pattern Hopfield model 的稳定-retrieval criterion, 不是大脑记忆容量结论。

11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. asynchronous update 是什么?

2. energy descent 需要哪些 weight 条件?

3. 为什么 $\Delta E \leq 0$?

4. Hebbian W_{ij} 存了什么?

5. $P=1$ 时 ξ 为什么 fixed point?

6. $-\xi$ 为什么也 attractor?

7. overlap $m=1$ 、0、 -1 分别表示?

8. crosstalk mean/variance 近似?

9. capacity 为什么 $\propto N$?

10. odd mixture state 是什么?

11. $\alpha_c \approx 0.138$ 的适用范围?

12. sparse model 为什么可能更高效?

13. 公式表中的“binary neuron state”解决什么问题? 写出适用条件与一个 sanity check。

14. 公式表中的“asynchronous update”解决什么问题? 写出适用条件与一个 sanity check。

15. 公式表中的“Hopfield energy”解决什么问题? 写出适用条件与一个 sanity check。

16. 公式表中的 “Hebbian storage rule” 解决什么问题？写出适用条件与一个 sanity check。

17. 公式表中的 “pattern overlap” 解决什么问题？写出适用条件与一个 sanity check。

18. 公式表中的 “retrieval field decomposition” 解决什么问题？写出适用条件与一个 sanity check。

19. 公式表中的 “loading ratio” 解决什么问题？写出适用条件与一个 sanity check。

20. 公式表中的 “classic capacity threshold” 解决什么问题？写出适用条件与一个 sanity check。

12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
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12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“binary neuron state”对应的变量关系入手。
Hint 2. 把条件“classic dense model”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“asynchronous update”对应的变量关系入手。
Hint 2. 把条件“one neuron at a time”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “Hopfield energy” 对应的变量关系入手。
Hint 2. 把条件 “symmetric W” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “Hebbian storage rule” 对应的变量关系入手。
Hint 2. 把条件 “ $W_{ii}=0$ ” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “pattern overlap” 对应的变量关系入手。
Hint 2. 把条件 “ ± 1 patterns” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
18. **Hint 1.** 先从 “retrieval field decomposition” 对应的变量关系入手。
Hint 2. 把条件 “random-pattern analysis” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
19. **Hint 1.** 先从 “loading ratio” 对应的变量关系入手。
Hint 2. 把条件 “P memories, N neurons” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
20. **Hint 1.** 先从 “classic capacity threshold” 对应的变量关系入手。
Hint 2. 把条件 “dense unbiased random Hopfield” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。

13 Complete Answer Key with reasoning

1. 随机/依序选择一个 neuron, 用 local field 更新, 再继续。
2. W symmetric, 通常 $W_{ii}=0$ 。
3. new state 与 local field 对齐, 使 $-\Delta S_i h_i \leq 0$ 。
4. 所有 patterns 中 bits i, j 同号/异号的 summed correlation。
5. local field $= (N-1)\xi_i/N$, 与 ξ_i 同号。
6. global sign flip 不改变 pair products/energy。
7. 完全同 pattern、平均无关、完全反号。
8. mean 0, variance $\approx P/N = \alpha$ 。
9. signal $O(1)$, crosstalk variance P/N ; 保持 finite 需 $P=O(N)$ 。
10. 若干 memories 带符号求和后取 sign 的 spurious attractor。
11. 经典 dense random unbiased asynchronous Hopfield large- N model。
12. 低 activity 与 global inhibition 减少 interference, 并按 information per synapse 优化。
13. 适用条件/约定: classic dense model. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$S_i \in \{-1, +1\}$$

14. 适用条件/约定: one neuron at a time. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$S_i \leftarrow \text{sgn}(h_i)$$

15. 适用条件/约定: symmetric W . sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$E = -\frac{1}{2} \sum_{ij} W_{ij} S_i S_j$$

16. 适用条件/约定: $W_{ii}=0$. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$W_{ij} = N^{-1} \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}$$

17. 适用条件/约定: ± 1 patterns. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$m^{\mu} = N^{-1} \sum_i \xi_i^{\mu} S_i$$

18. 适用条件/约定: random-pattern analysis. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$h = \text{signal} + \text{crosstalk}$$

19. 适用条件/约定: P memories, N neurons. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\alpha = P/N$$

20. 适用条件/约定: dense unbiased random Hopfield. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\alpha_c \approx 0.138$$

14 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture26-Memory-HopfieldModel.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture26-Memory-HopfieldModel.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture26-Memory-HopfieldModel.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture26-Memory-HopfieldModel.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture26-Memory-HopfieldModel.pdf	5	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture26-Memory-HopfieldModel.pdf	6	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.

15 Errata, uncertainty log, and external endnotes

15.1 Errata / source cautions

1. Page 2 的 Nobel 旁注是历史背景，不影响模型定义。
2. Page 5 的 $\alpha_c \approx 0.138$ 仅属于 classic dense unbiased random-pattern Hopfield retrieval criterion.
3. Page 6 的 sparse stochastic model 是 alternative family, capacity comparison 依 activity sparsity 与 information metric.

15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.